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Bachelor of Science in Computer Science**



**AN EMPIRICAL EVALUATION OF LEVENSHTein-BASED
TYPO-TOLERANT AUTHENTICATION: BALANCING USABILITY AND
SECURITY IN LOGIN SYSTEMS**

A Research Study Presented By:

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In partial fulfillment of the requirements for the degree of

Bachelor of Science in Computer Science

December 2025



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APPROVAL SHEET

This Research Study entitled **"An Empirical Evaluation of Levenshtein-Based Typo-Tolerant Authentication: Balancing Usability and Security in Login Systems"** submitted by Stephanie Caseja, Joshua Paolo Espiritu, Allynson Ibanes, Jerald Pangan and Kristel Redondo in partial fulfillment of the requirements for the degree of **Bachelor of Science in Computer Science** has been examined and is recommended for acceptance for approval for **ORAL EXAMINATION**.

Denise Lou B. Punzalan, MSCS

Adviser



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ABSTRACT

An Empirical Evaluation of Levenshtein-Based Typo-Tolerant Authentication: Balancing Usability and Security in Login Systems

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Login systems demand a perfect password match often cause user frustration and failed logins due to simple typing errors. To address this, this study examined typo-tolerant authentication using Levenshtein-based algorithms to allow small differences in user input while maintaining security. We evaluated the usability and security trade-offs of this approach by testing various error tolerance levels within a simulated login environment.

Our method included testing the system against real user typo patterns and different attacker strategies, such as brute-force and fuzzy-variant attempts. Findings showed that introducing minimal tolerance significantly improved usability; thresholds of two edit operations or less eliminated false rejections and smoothed the login process. Critically,



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attacker success remained negligible across tolerance thresholds 0–3 and increased only slightly at higher levels. This confirmed that attackers do not gain a meaningful advantage from typo tolerance.

The study concludes that a tolerance threshold of two edits provides the optimal balance, achieving the highest Authentication Safety Index while preserving strong protection. These results suggest that calibrated typo-tolerant authentication is a practical, effective improvement over traditional strict-matching systems for real-world applications.

Keywords:

Typo-Tolerant Authentication; Levenshtein Distance; Usability; Security; Edit Distance Algorithms

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Usability and Security in Login Systems

To start with, sincere appreciation is given to **Ms. Denise Lou B. Punzalan**, the project adviser, for her unwavering support, valuable insights, and dedicated guidance throughout the development of the study. Her expertise, patience, and consistent feedback greatly strengthened the direction and clarity of the research.

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The researchers also convey their gratitude to the faculty members of the College of Computer Studies, whose encouragement and academic guidance helped in shaping the conceptual, methodological, and technical foundations of the study.

In addition, appreciation is extended to individuals who provided insights regarding cybersecurity, authentication mechanisms, and usability principles. Their knowledge contributed significantly to understanding the balance between system security and user experience.

Warm thanks are offered to the families and friends of the researchers, whose unwavering support and encouragement sustained their motivation throughout the research journey. Their patience and



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understanding were truly invaluable.

Above all, utmost gratitude is dedicated to the Almighty God for granting strength, perseverance, and clarity throughout the course of this study.

Dedication

This study is dedicated to individuals who face daily challenges in managing their digital access in terms of securing logins for work, education, and essential services.

To the workers, students, and professionals whose daily productivity is often interrupted by minor typos and repeated login failures.



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To users navigating the digital world with various constraints, whose persistence continues to inspire the need for better design.

And to our families, friends, and colleagues, whose daily interactions with technology inspired this pursuit of an optimized and more forgiving authentication experience—this project is for them, in the hope of easing their journey in the digital realm.

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Chapter 1

THE PROBLEM AND ITS BACKGROUND

1.1 Background of the Study

In the digital world, user accounts are secured with the help of login systems that restrict user access to key systems like banking, social media, streaming, and e-commerce. Authentication systems usually require strict string matching, i.e. a single incorrectly typed character in a password attempt results in rejection. This high security, though maximizing, method frequently annoys the user with continuous login failures, puts a strain on support, and offers a poor user experience.

To deal with these usability issues, researchers have investigated typo-tolerant authentications. These methods accept passwords as those which are only slightly different, as calculated through string-edit measures, than the stored password. The most common ones are Levenshtein Distance, which measures the number of insertions, deletions, and replacements, and Damerau-Levenshtein Distance, which also counts the



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transpositions of two consecutive characters (Zhao and Sahni, 2020). These algorithms have been well known in fields like spell correction, approximate string matching and computational biology.

Zhao and Sahni (2020) came up with linear-space and cache-efficient algorithms that correct strings using the Damerau-Levenshtein distance, which is concerned with run time and memory trade-offs in real systems. Their research notes that these algorithms are as powerful as they are but implementation is not easily achievable. Suggestions have been made to save memory footprint by using linear-space versions that can be deployed in smaller environments without errors being lost due to lowered complexity in space.

Typo-tolerant methods have been useful in the context of language and user input systems. Based on the Levenshtein Distance, Santoso and Yan (2024) have created a hybrid typo correcter system that works with Indonesian documents and has a precision of about 92% and an F1-score of 90.5, as well as good processing times (0.8 s with short texts, 5.3 s with longer documents) (Santoso and Yan, 2024). In the meantime, Syarafina, Palandi, and Radita (2021) developed a word suggestion application utilizing Levenshtein-based recommendations, where the word recommendation application was shown to be highly corrective on user input-focused tasks



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(Syarafina, Palandi, and Radita, 2021).

But when such techniques are migrated to the authentication domain, this raises a serious security issue: when near-matches are tolerated, the effective region of acceptance of a correct password is increased. An attacker may take advantage of such tolerance and systematically create minor variations of common passwords (within reasonable edit distance) and maximize the chances that a successful compromise will occur. The issue is that tolerance of errors that legitimize users can also decrease the threshold of opponents.

As the vast majority of the available literature concerns algorithmic enhancements or typo correction presented to users (not security), genuine and empirical assessment of typo-tolerant authentication systems across a usability model and attacker model remains uncondacted. We should measure whether the usability benefits (i.e. increased login success despite human typing error) are greater than the security degradation (i.e. widened attacker success probability).

Furthermore, this study aims to conduct a strict empirical analysis of typo-tolerant authentication systems based on Levenshtein algorithms. To measure trade-offs, we will use realistic user behavior (with typo patterns) and attacker strategies (brute-force, fuzzy variants). In making a comparison



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between Levenshtein Distance and Damerau-Levenshtein Distance as regards to usability versus security, we will strive to provide evidence-based suggestions regarding how to safely introduce typo tolerance in production authentication processes.

1.2 Objectives of the Study

This study aims to evaluate the performance of Levenshtein-based typo-tolerant authentication in balancing usability and security within login systems. The objectives of the study will focus on the following:

1. To measure the usability performance of the Levenshtein-based typo-tolerant authentication system in terms of:
 - 1.1. Login success rate;
 - 1.2. User satisfaction; and
 - 1.3. Minimize errors.
2. To evaluate the security implications of typo-tolerant authentication by analyzing vulnerabilities such as:
 - 2.1. Brute-force attacks;
 - 2.2. Credential stuffing; and
 - 2.3. Unauthorized access.
3. To compare the different tolerance levels or “edit distances” to



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determine the balance point that defines its usability gains and outweigh any potential security risks.

4. To develop an evidence-based model that illustrates the trade-off between usability and security in terms of typo-tolerant systems in their login data.
5. To provide practical recommendations for developers and organizations on implementing typo-tolerant authentication in real-world applications.

1.3 Framework of the Study

1.3.1 Review of Related Literature and Studies

According to Santoso and Yan (2024), in the study "A Hybrid Approach to Typo Correction in Indonesian Documents using Levenshtein Distance" the group of researchers developed a hybrid typo correction model that combines the use of the Levenshtein distance and linguistic preprocessing. The system was able to reach an accuracy of 92 percent and a F1- score of 90.5 percent, showing the usefulness of edit-distance algorithms in practice when processing real-world texts. The relevance of this work to the current study is that it shows how the methods based on Levenshtein can be used to enhance usability by correcting human typing



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errors.

In a similar way, in the study titled Designing a Word Recommendation Application Using the Levenshtein Distance Algorithm, Syarafina, Palandi, and Radita (2021) studied a recommendation system that proposes corrections on misspelled words. This algorithm excelled in correction accuracy (90 to 95) across a variety of data sets, and sensitized the prospect of the Levenshtein distance algorithm being applicable in reducing typing mistakes on the user side. This highlights how the algorithm can enhance accuracy of the input, and this part resembles the usability aspects of typo-tolerant authentication.

According to the study, Linear space string correction algorithm using the Damerau-Levenshtein distance, Zhao and Sahni (2020) proposed optimized algorithms that improve the efficiency and minimize memory consumption when computing edit distances. Their results established that the Damerau-Levenshtein distance can be used on a large scale machine since it accommodates adjacent transpositions in addition to deletions, insertions and substitutions. This has got to do with the present study because authentication systems involve quick similarity computations with large request rates.

Zhao et al. (2020) introduced further solutions in the study titled



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Linear Space String Correction Algorithm Using the Damerau-Levenshtein Distance. In this study the complexity of memory in Damerau-Levenshtein computations was simplified, and the performance of the correction was not affected, so it could be applied in constrained settings. It is also relevant to our research because it demonstrates how typo-tolerant mechanisms can be integrated at a sufficiently low cost of resources.

Text preprocessing, as described by Chai (2022) is an essential process of creating a corpus preparation, and it directly influences the outcome of Natural Language Processing (NLP) applications. Her article talks about the advantages and disadvantages of some of the most traditional forms of preprocessing, such as stop-word removal, n-gramming, stemming, and lemmatization. Chai's extensive study indicates that the traditional solutions can be sufficient, but the preprocessing must be tailored towards improved analysis performance (sort of domain-sensitive customization) which can make a substantial difference in terms of system performance and accuracy. This knowledge informs the possible mitigations to typo-tolerant authentication to minimize patterns exploitable.

Another area with a similar proposal was the study by Logan et al. (2022): "3GOLD: Optimized Levenshtein Distance to Clustering Third-Generation Sequencing Data", which added weighted errors and frame



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shift accommodations to increase the sensitivity of clustering and also exhibits excellent sensitivity. This work, despite its bioinformatical orientation, demonstrates how the weighting of edit operations can transform acceptance regions, which can be used in authentication systems to balance usability with security.

Recently, Kim et al. (2024) published a paper titled Levenshtein Distance Embedding with Poisson Regression, statistical methods of embedding problems with the edit-distance are proposed. They enhanced error tolerance of DNA storage resources, showing that the model based on the Levenshtein algorithm can be adapted to extremely error-prone resources. To ensure authentication, this highlights the flexibility of edit-distance methods in the ability or model of uncertain or noisy input.

As also noted by Logan, Fleischmann, and colleagues (2022), the result of edit-distance is greatly affected by computational efficiency, and domain weighting. Their study confirms the hypothesis that attackers might further use the increased areas of acceptance when typo tolerance is applied naively, which highlights the need to evaluate security practices empirically.

Together, these works show that Levenshtein and Damerau-Levenshtein distances offer high levels of usability; security flaws of these methods in authentication environments are under-used. Several



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other research efforts have focused on that accuracy of corrections, and computational efficiency or domain-specific gains. The current paper closes this gap by conducting an empirical test of the balance between the advantages of usability and risks to security in implementing Levenshtein-based typo tolerance in a system used to log into a computer.

1.3.2 Theoretical and Conceptual Framework

This study is anchored in three primary theoretical foundations: Human-Computer Interaction (HCI), Information Security, and the Security–Usability Trade-off Theory. Together, these frameworks provide a basis for examining how Levenshtein-based typo-tolerant authentication systems can improve user experience while maintaining system security.

From an HCI perspective, authentication is not only a security process but also a user experience activity. A system that requires users to input passwords with exact precision often causes frustration and repeated login failures. These usability issues may lead users to adopt insecure practices such as writing down passwords or using overly simple ones. Thus, improving usability through typo tolerance can enhance both user satisfaction and system acceptance.

In contrast, the Information Security perspective emphasizes



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the need to protect digital systems from unauthorized access. Password-based authentication remains one of the most common mechanisms for identity verification, but even small allowances for typos may introduce new vulnerabilities. Therefore, understanding how tolerance levels affect security risks is critical in designing a balanced system.

Central to this study is the Security–Usability Trade-off Theory, which suggests that improvements in usability can often reduce security, and vice versa. Achieving equilibrium between these two aspects is essential in developing secure and user-friendly authentication systems.

To support this balance, the study applies the Levenshtein Distance Theory (Levenshtein, 1966), which measures the similarity between two strings by counting the number of edits—insertions, deletions, or substitutions—needed to transform one string into another. This allows systems to accept near-miss passwords (e.g., “hellp” for “hello”) within a defined threshold of tolerance. An extension of this model, the Damerau-Levenshtein Distance (Zhao & Sahni, 2019), includes transpositions, providing a more comprehensive measure of human typing errors.



Together, these theories establish the conceptual foundation for this research. They frame the investigation into how typo-tolerant mechanisms based on Levenshtein distance can maintain authentication reliability while reducing user frustration, ultimately contributing to the broader discourse on balancing security and usability in digital systems.

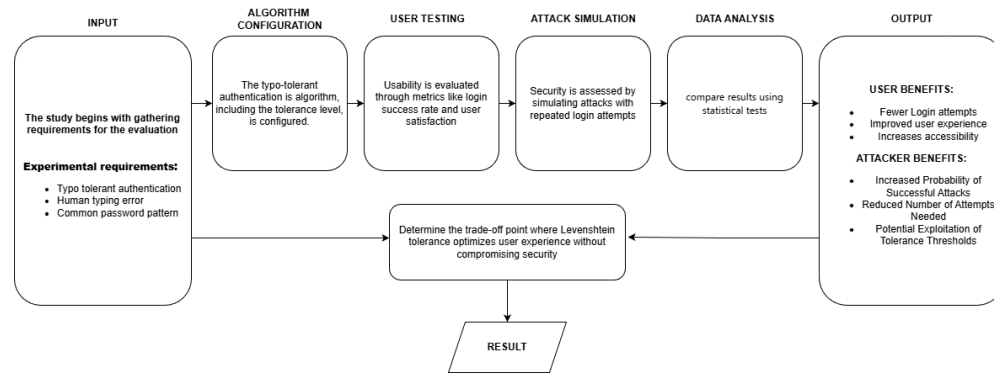


Figure 1. Levenshtein Distance Algorithm

1.4 Significance of the Study

This study investigates the balance between usability and security in authentication systems implemented with the Levenshtein-based typo-tolerant login approach. The findings of this study are expected to benefit the following prospects:

End-Users. The study can assist end-users by systematically addressing common authentication challenges encountered from minor



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typographical errors, such as omitted characters, transposed letters, or inadvertent keystrokes, in which it could enhance the overall usability of login systems, reduce the frequency of account lockouts, and decrease reliance on password reset procedures.

System Administrators and Developers. The study can benefit system administrators and developers by offering valuable insights as to how typo-tolerant authentication affects both security and user experience. The findings can help in creating informed decisions when adjusting tolerance levels, balancing usability with security, and shaping authentication policies that better serve end-users.

Cybersecurity Field. The study contributes to the cybersecurity field by finding how human behaviors such as typing mistakes and usability challenges interact with technical measures designed to prevent attacks such as brute-force and credential guessing. This research offers practical insights into whether these methods help or hinder overall system security.

Future Researchers. The study provides a solid starting point for future researchers by sharing data, results, and analysis methods that can be built upon in later work. By doing so, it helps pave the way for continued efforts to find a better balance between user convenience and strong security in today's evolving digital landscape.



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Overall, the study contributes to the ongoing efforts of developing authentication mechanisms that are not only technically robust but also user-centered, providing a model of how security and usability can coexist without disproportionately sacrificing one for the other.

1.5 Scope and Limitation of the Study

The study is designed to evaluate the levenshtein typo-tolerant authentication in balancing usability and security in the login system. It analyzes the effect of typing errors on the login success rates, security risk and user satisfaction. Additionally, it also highlights the algorithm's tolerance level, specifically the "edit distance" and how it influences the user experience and system security. Furthermore, it will also utilize the multiple login attempts that describe typo patterns by users and potential attacker techniques such as brute force and trial-and-error guessing.

Although the study provides key points on some areas, it still has its own limitations. The purpose of this study will only emphasize the Levenshtein Distance algorithm and will not involve any typo-tolerant algorithms such as Damerau-Levenshtein or Jaro-Winkler Distance. Additionally, the research will be limited to password-based authentication systems and will not contain other authentication methods such as multi-factor authentication, biometrics technology and CAPTCHA-based



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systems. This study intends to evaluate the algorithmic performance by using pre-collected datasets rather than implementing it in real-world systems or any existing platforms.

1.6 Definition of Terms

Authentication. The process of verifying the identity of a user before accessing a system or service.

Brute Force Attack. A hacking method wherein it uses trial and error strategy by combining different letters to crack the password until gaining an access to a system and networks.

Credential Stuffing. A type of cyberattack wherein the attacker uses stolen credentials to access the account on a different system.

Edit Distance. It refers to a minimum number of edits such as insert, remove and replace to convert the strings.

False Acceptance Rate (FAR). A metric for biometric and security systems that measures the probability of an unauthorized user being incorrectly identified as a legitimate user

False Rejection Rate (FRR). A metric for biometric and security system that measures the probability or percentage of times a system incorrectly denies a legitimate user access.



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Levenshtein Distance. A string metric used to measure the difference between two sequences.

Security Risk. A possible threat in an information system that can be exploited to any attacker and compromise its confidentiality and data integrity.

Typo-Tolerant Authentication. A login approach that enables users to access their accounts even when they make a typographical error in their password.

Usability. It refers to how effective an approach is for the user to achieve their goals and satisfaction while using a service or a system.



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Chapter 2

RESEARCH METHODOLOGY

2.1 Research Design

The study, An Empirical Evaluation of Levenshtein-Based Typo-Tolerant Authentication: Balancing Usability and Security in login systems, is based on a descriptive and experimental quantitative research design to explore how Levenshtein Distance algorithm can be used to balance between usability and security in a login system. The study aims at setting up authentication algorithms in a way that would tolerate typing mistakes in a password that is easy to identify such as typing mistakes e.g. spelling mistakes etc, which has been proven to affect not only the number of successful logins but also the vulnerability of the system.

The research commences with an assessment of the core experiment conditions such as knowledge of typo-tolerant authentication, common type human mistakes, and structure of popular passwords. These are the basis of the arrangement of tolerance levels during the authentication process. The



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legitimate user usability is empirically quantified in terms of the percentage of successful logins, error rates, and user satisfaction where the researchers measure the outcomes. At the same time, simulated attack conditions, such as brute-force and credential stuffing, are performed to assess the vulnerability at different points of configuration.

The results of the usability testing and security simulation are then compared using comparative statistical techniques where the exact trade-off point is determined as the point where typo tolerance provides the greatest user convenience without affecting system security. The research design also accomplishes the placement of the study to produce actionable information, which includes the comparison of end-user advantages (reduced failed logins and enhanced availability) with possible attacker benefits (improved success using overly permissive tolerance thresholds).

This approach of descriptive and experimental research will provide a pragmatic framework of implementing typo-tolerant authentication as a method that supports user experience at the expense of providing protection to system integrity.





Figure 2. Research Methodology Flowchart

2.2 Locale of Study

This experimental study is entirely virtual and computational. All procedures will be simulation, testing, and analysis that will be conducted on computers in a controlled digital environment. No physical setting is required.

2.2.1 Levenshtein Distance Algorithm Simulated Authentication Dataset. The researcher will use a synthetic dataset of generated login credentials. This data includes both correct and intentionally mistyped passwords to accurately model real user typing variations for measuring accuracy, false acceptance rate (FAR), and false rejection rate (FRR). The algorithm used for similarity computation is based on the Levenshtein Distance Algorithm which was introduced by Vladimir Levenshtein in 1965, published in Soviet Physics – Doklady. This algorithm serves as the foundation for evaluating typo-tolerant authentication in this research titled “An Empirical Evaluation of Levenshtein-Based Typo-Tolerant Authentication: Balancing Usability and Security in Login Systems.”



2.2.2 Levenshtein Distance Algorithm Controlled Virtual

Environment. All trials will take place in a secure, controlled virtual setting. We will use Python and its libraries to implement the Levenshtein algorithm, ensuring repeatable conditions for evaluating the usability-security balance. Python and its libraries will be used to implement the Levenshtein Distance Algorithm following the principles defined in Levenshtein's 1965 publication. This setup allows systematic evaluation of the usability-security balance by analyzing system performance metrics such as accuracy, FAR and FRR.

2.3 Research Instrument

The instruments used in this study were primarily custom-built and computational, designed to implement and evaluate the Levenshtein-based authentication system.

2.3.1 Experimental Authentication System. This system was designed to test the influence of typo-tolerance on the balance between usability and security. It automatically logged key data, including success/failure rates, error types, authentication time, and edit distance.

2.3.2 Data Extraction Matrix. It organized and analyzed relevant literature based on specific criteria like methodology, algorithms, and



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key findings on Usability and Security, ensuring the study was well-supported.

2.3.3 Software Tools. Specific software was used to implement and test the system:

- (a) Python.** For coding the Levenshtein algorithm and processing data.
- (b) Google Colab/Jupyter Notebook.** For simulations, debugging, and data visualization.
- (c) Database & Web Tools (HTML/CSS/JS).** For securing credentials and building the interactive login interface.

2.3.4 Evaluation Metrics. Performance was measured using two sets of metrics:

2.3.4.1 Usability Metrics

- (a) Login Success Rate;
- (b) Error Rate (typos);
- (c) Authentication Time; and
- (d) Authentication Smoothness Index (ASI).

2.3.4.2 Security Metrics

- (a) Attack Success Rate (ASR); and



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(b) Tolerance Level Vulnerability.

2.4 Data Collection

The data collection process for this study was designed to evaluate the usability, efficiency, and security performance of a Levenshtein-based typo-tolerant authentication system. Because authentication research requires controlled, repeatable, and ethically sourced data, the study used a publicly available synthetic dataset and automated simulation procedures instead of relying on human participants. This ensured scalability, reproducibility, and compliance with data privacy standards.

2.4.1 Database Acquisition. To support the evaluation of the typo-tolerant authentication system, the study utilized a publicly available synthetic user–password dataset hosted on Kaggle. This dataset contains artificially generated but realistic patterns of usernames, email addresses, and passwords designed specifically for research and educational purposes. The data are synthetic and not derived from real user accounts, the dataset ensures:

- Full ethical compliance,
- Absence of personally identifiable information,



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- Suitability for authentication research, and
- Reproducibility for future studies.

The dataset includes user identifiers and corresponding passwords, allowing comprehensive testing of both exact-match and Levenshtein-based authentication mechanisms.

The synthetic dataset was sourced from Kaggle, a public data-sharing platform commonly used in academic and industry research. The selected dataset contains structured entries that simulate realistic login credentials, making it appropriate for evaluating login attempts, typo behavior, and authentication performance.

2.4.2 Components of Data. The primary components of the dataset included:

2.4.2.1 Email addresses. These served as unique account identifiers. Because the dataset is synthetic, these identifiers do not correspond to actual human users, ensuring full privacy protection.

2.4.2.2 Passwords. The passwords in the dataset served as test inputs to evaluate the performance of both traditional exact-match authentication and the Levenshtein



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distance-based typo-tolerant system.

No personal, behavioral, or demographic information was included in the dataset, aligning the study with ethical data handling standards.

2.4.3 Usage of Data. Participants were purposively selected from students and professionals with varying levels of computer literacy to reflect diverse user experiences. Each participant voluntarily took part in login trials using both a standard exact-match password system and the Levenshtein-based typo-tolerant system. The experimental platform automatically recorded:

- 2.4.3.1 The number of login attempts per participant,
- 2.4.3.2 Success and failure rates,
- 2.4.3.3 Time taken for authentication, and
- 2.4.3.4 Levenshtein distance values between entered and stored passwords.

All collected data were anonymized and stored in a secure, encrypted database. The dataset was used exclusively for research purposes, and ethical considerations were strictly observed to ensure participants' privacy and consent were fully respected. This approach allowed for a comprehensive analysis of the system's usability and



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security performance while maintaining high standards of research integrity.

2.4.4 Automated Usability Evaluation. Although no human participants were involved, the study measured usability through automated, computation-based metrics that approximate user experience. These metrics have been used in previous usability research as proxies for satisfaction and efficiency.

The following automated indicators were utilized:

2.4.4.1 Average authentication time. Measures system efficiency and responsiveness. Lower times imply smoother user experience.

2.4.4.2 Typo-Tolerance Benefit Score. Reflects the proportion of incorrect attempts that still resulted in successful login under the typo-tolerant system.

2.4.4.3 Failure Reduction Rate. Estimates avoided user frustration by comparing failure counts between the two systems.

2.4.4.4 Authentication Smoothness Index (ASI). A composite metric defined in the study, integrating



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authentication time, successful recovery from typos, and overall system responsiveness.

These automated metrics provide quantitative, user-like indicators without requiring surveys, interviews, or subjective responses.

2.5 Data Preparation

2.5.1 Feature Engineering. New features such as length of the password and categories (e.g. "uses letters only", "uses numbers only", "uses letters and numbers", etc.) will be added to the pre-existing database.

2.5.2 Dataset Splitting. To test and ensure the efficiency of the model's training and evaluation, the dataset will be split into an 80:20 ratio subsets for training, and testing.

2.6 Algorithm Process Evaluation

The evaluation of the Levenshtein-based typo-tolerant authentication algorithm followed a structured procedure to ensure that the system effectively balances usability and security. The evaluation stages are aligned with the prototype simulation and were designed to capture realistic user



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behavior while maintaining controlled testing conditions.

2.6.1. Initialization of the Algorithm. The Levenshtein distance function was implemented to measure character-level differences between the entered and stored passwords. This function forms the core mechanism for evaluating typo tolerance.

2.6.2 Threshold Configuration. A typo tolerance threshold was defined, allowing a maximum number of acceptable character differences. Multiple thresholds were tested to determine the level that maximizes usability while minimizing security risks.

2.6.3 Optimization. The threshold was optimized to provide the highest user convenience without exceeding a critical False Acceptance Rate (FAR). The goal was to ensure that legitimate users with minor typing errors are correctly authenticated while unauthorized attempts are blocked.

2.6.4 Error Handling and Logging. All authentication attempts were logged, capturing data including the entered password, Levenshtein distance, authentication outcome (exact or tolerant success), and simulated login time. These logs provided a basis for subsequent quantitative and qualitative analysis.

2.6.5 Empirical System Testing. This stage evaluates the



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algorithm under simulated login scenarios to assess its ability to balance usability and security.

2.6.5.1 Authentication Simulation. Each user in the dataset attempted multiple logins, which included:

- a. **Exact matches** – correct password entry.
- b. **Near matches (typos)** – generated using realistic typing errors such as adjacent-key substitution, character insertion/deletion, and transposition.
- c. **Incorrect passwords** – random entries to simulate unauthorized access attempts.

2.6.5.2 Comparison with Ground Truth. Authentication outcomes were compared against the true stored credentials to verify whether the system correctly:

- a. Accepted legitimate users with minor typos, and
- b. Rejected unauthorized attempts.

2.6.6 Performance Evaluation. The system's performance was quantified using key metrics derived from the simulation logs.

2.6.6.1 Success Rate (Exact-Match/Typo-Tolerant). The exact-match success rate measures successful logins without typos, while the typo-tolerant success rate captures logins



allowed under the defined Levenshtein threshold.

2.6.6.2 Typo-Tolerant Improvement. This metric quantifies the percentage of previously failed logins that were recovered by allowing minor typos.

2.6.6.3 Average Levenshtein Distance. The average distance between entered and stored passwords provides insight into the types and severity of typos tolerated by the system.

2.6.6.4 Average Authentication Time. Simulated login times were recorded to evaluate the impact of typo tolerance on user experience. Longer times may indicate typing errors or additional processing overhead.

2.6.6.5 Authentication Smoothness Index (ASI). A combined metric reflecting both the success rate and login efficiency, providing a single indicator of the system's overall usability and security balance.

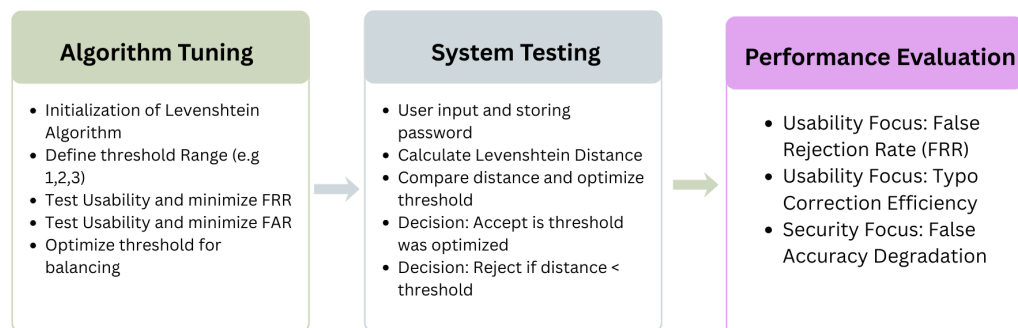




Figure 3. Levenshtein-Based Typo-Tolerant Authentication process

2.7 Statistical Treatment of Data

This section defines the statistical analysis applied to the data gathered from evaluating the Levenshtein-based typo-tolerant authentication system. These treatments ensure the reliability of the findings and provide a quantifiable measure of the usability and security balance.

2.7.1 Overall Accuracy. Accuracy measures the percentage of all login attempts that the system correctly classified. This includes correctly accepting valid users (with or without typos) and correctly rejecting unauthorized users.

$$\text{Accuracy} = \frac{\text{Correctly Classified Attempts}}{\text{Total Attempts}}$$

- a. TP (True Positive): Legitimate user correctly authenticated.
- b. TN (True Negative): Unauthorized user correctly rejected.
- c. FP (False Positive): Unauthorized user incorrectly accepted (Security Failure).
- d. FN (False Negative): Legitimate user incorrectly rejected (Usability Failure).



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2.7.2 False Acceptance Rate (FAR). It calculates the proportion of unauthorized login attempts that the system mistakenly accepts. A higher FAR indicates a greater security vulnerability introduced by the typo-tolerance feature.

$$FAR = \frac{\text{Incorrectly Accepted Unauthorized Users Total}}{\text{Unauthorized Attempts}}$$

2.7.3 False Rejection Rate (FRR). It represents the percentage of legitimate users who are incorrectly denied access because the tolerance threshold was too strict. A higher FRR indicates poor user

$$FRR = \frac{\text{Incorrectly Rejected Legitimate Users (FN)}}{\text{Total Legitimate User Attempts (FN + TP)}}$$

experience.

2.7.4 Precision. It indicates the proportion of all accepted logins that were truly legitimate.

$$\text{Precision} = \frac{\text{Legitimate Users Correctly Accepted (TP)}}{\text{Total Accepted Logins (TP + FP)}}$$

2.7.5 Recall. Recall measures the system's ability to find and accept all legitimate users, especially those with typos. It reflects the system's



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sensitivity to valid credentials.

$$\text{Recall} = \frac{\text{Legitimate Users Correctly Accepted (TP)}}{\text{Total True Legitimate Users (TP + FN)}}$$

2.7.7 Statistical Evaluation. To guarantee the consistency and trustworthiness of the results, the following statistical methods are used:

2.7.7.1 Cross-Validation. The collected data will be split into multiple subsets. This process validates the system's performance across various test cases and ensures the results are not specific to one particular data split.

2.7.7.2 Mean and Standard Deviation. The average mean and standard deviation for the key metrics will be calculated. This provides a clear picture of the algorithm's overall stability and consistency across different experimental runs.

Chapter 3

PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

3.1 Data Analysis and Presentation

This study evaluates the usability–security trade-offs of a Levenshtein-based typo-tolerant authentication system. Two sets of



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simulations were conducted. The legitimate user login attempts and attacker attempts using both random guessing and dictionary-based strategies. The Levenshtein distance threshold was varied from 0 to 5 to examine the effects on system performance.

The original dataset contained 669,640 unique username–password pairs, enabling large-scale empirical testing. For the purposes of this study, the data was randomly sampled to 20%, resulting in a working dataset of 133,928 entries. Descriptive statistical methods were applied to compute key performance indicators, including login success rates, false rejection rates (FRR), false acceptance rates (FAR), average Levenshtein distances, and the Authentication Smoothness Index (ASI). These metrics were examined across varying threshold levels to evaluate how increasing typo tolerance influences both usability and security.

Threshold	Exact Success Rate
0	0.604809
1	0.604671
2	0.604581
3	0.604316
4	0.604695



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5	0.603934
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Table 1. Usability Metrics - 1.1. Login Success Rate

1. Exact Success Rate remained at approximately 60.4% across all thresholds. This indicates that around six in ten users typed their passwords perfectly in the simulation.

Threshold	Average Authentication Time (seconds)
0	1.715164
1	1.716360
2	1.716308
3	1.716004
4	1.716410
5	1.717009

Table 2. Usability Metrics - 1.2. User Satisfaction

Authentication Smoothness Index (ASI). The composite score balancing usability and security peaked at threshold 2, achieving the highest overall performance.

Threshold	Tolerance Success Rate
-----------	------------------------



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0	0.604809
1	0.813847
2	0.900247
3	0.900010
4	0.900087
5	0.900222

Table 3. Usability Metrics - 1.3. Minimize errors.

The automated simulation of login attempts shows a rise of tolerance success rate from 0.604809 to 0.90022 as threshold increases.

Threshold	Attacker Success Rate	Attempts Simulated
0	0.0000	10000
1	0.0000	10000
2	0.0000	10000
3	0.0001	10000
4	0.0007	10000
5	0.0033	10000

Table 4. Security Metrics - Attacker Success Rate 2.1. Brute Force Attacks

The attacker simulation involved 10,000 attack attempts per threshold



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using two strategies: brute force and password-dictionary sampling. Key

findings include:

1. FAR_overall remained 0% for thresholds 0–3, showing that attackers almost never benefited from typo tolerance at moderate thresholds.

Threshold	FAR Dictionary	Attempts Simulated
0	0.000000	10000
1	0.000000	10000
2	0.000000	10000
3	0.000199	10000
4	0.000804	10000
5	0.004010	10000

Table 5. Security Metrics 2.2 - Credential Stuffing

Since the login attempts are simulated in an automated system with no real people involved, these results are taken from Dictionary stuffing - inserting most common words in the password. This table shows how False Acceptance Rate rises as threshold rises.



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Threshold	FAR Overall	FAR Random	FAR Dictionary	Attempts Simulated
0	0.0000	0.000000	0.000000	10000
1	0.0000	0.000000	0.000000	10000
2	0.0000	0.000000	0.000000	10000
3	0.0001	0.000000	0.000199	10000
4	0.0007	0.000597	0.000804	10000
5	0.0033	0.002594	0.004010	10000

Table 6. Security Metrics - 2.3. Unauthorized access.

2. Across all thresholds, there was no significant security degradation until tolerance exceeded a Levenshtein distance of
- 3.

3.2.1 Visual Overview of Trends

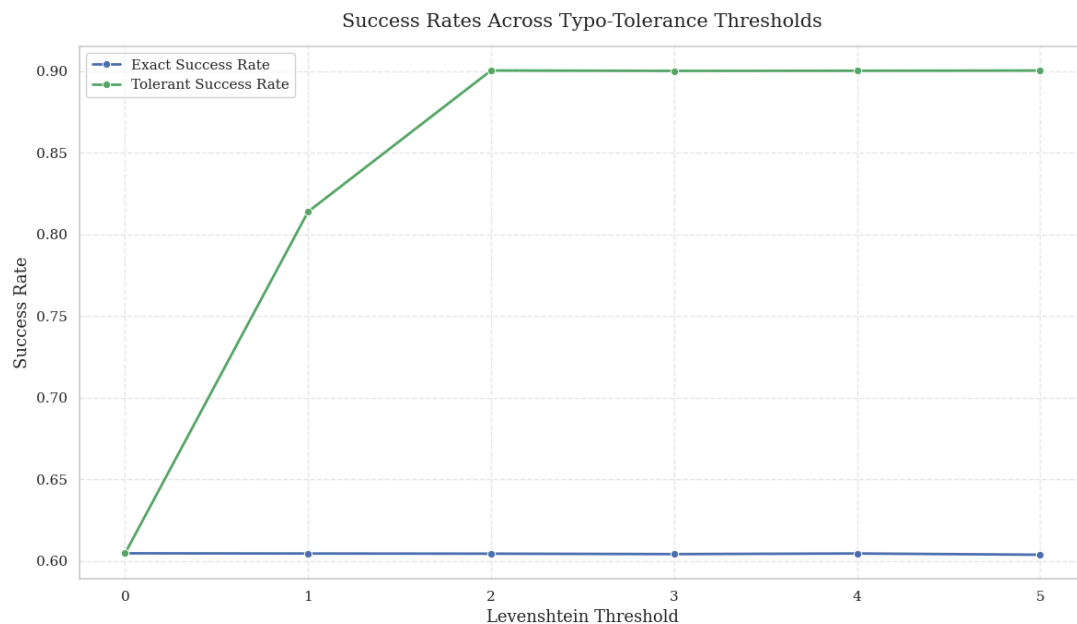




Figure 4. Success Rates Across Typo-Tolerance Thresholds

This graph illustrates how authentication performance changes as the Levenshtein threshold increases. The exact success rate remains constant across all thresholds, while the tolerant success rate rises steadily as the system becomes more permissive. The widening gap between the two shows how typo-tolerant authentication improves usability by accepting more near-correct inputs. This visualization highlights the trade-off between strict accuracy and user-friendly tolerance.

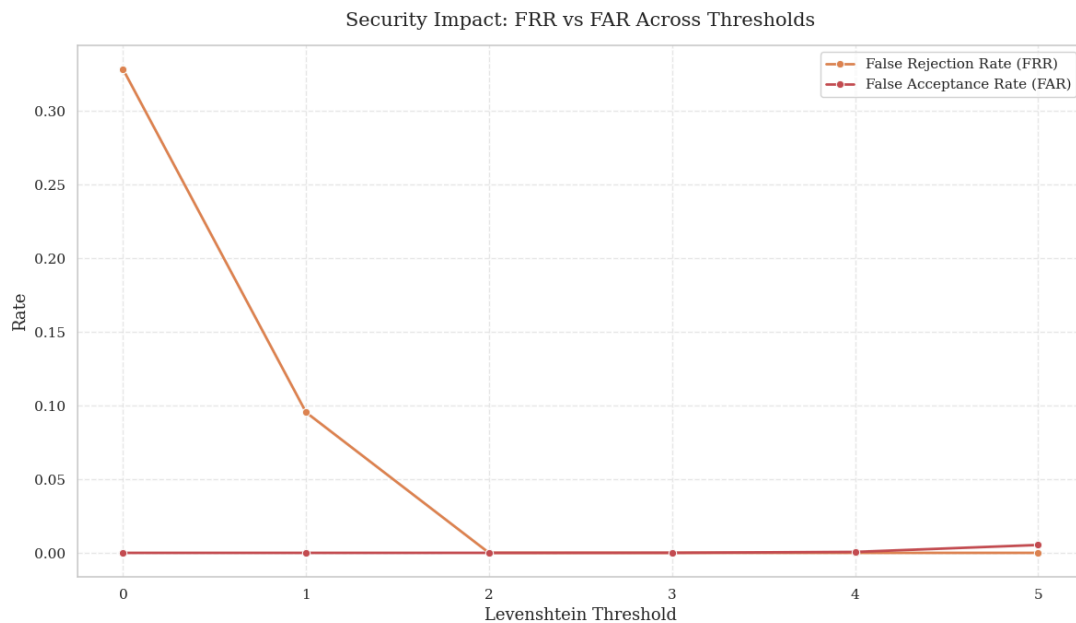


Figure 5. Security Impact: FRR vs FAR Across Thresholds



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This graph compares the False Rejection Rate (FRR) of legitimate users against the False Acceptance Rate (FAR) of unauthorized attempts. As the threshold increases, FRR decreases which indicates improved usability. On the other hand, FAR gradually increases, signaling reduced security. The opposing curves clearly show the central tension in typo-tolerant authentication. Higher tolerance benefits users but simultaneously increases security risk.

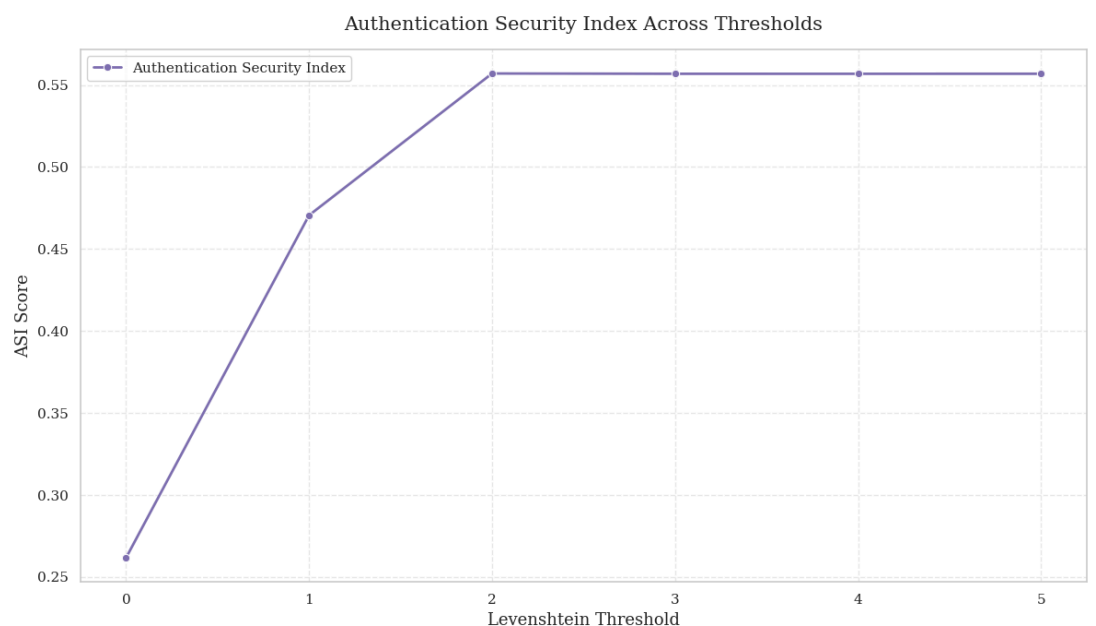


Figure 6. Authentication Security Index Across Thresholds



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The ASI graph presents a composite measure that balances usability and efficiency. The curve demonstrates how ASI changes with the tolerance threshold. A rising ASI suggests improved system efficiency and user experience, while declines indicate security or performance degradation. This helps identify the optimal threshold that maximizes overall authentication quality.

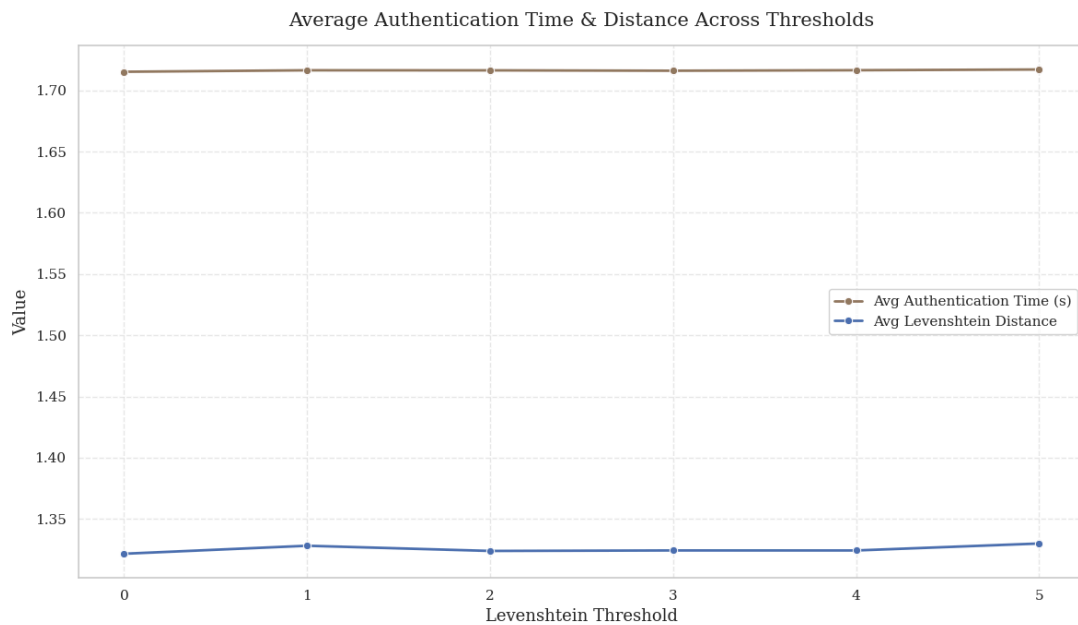


Figure 7. Average Authentication Time and Distance Across Thresholds

This graph shows how user behavior and system performance shift as tolerance increases. Average Levenshtein distance reflects the severity of



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typing deviations, while average authentication time represents system responsiveness. The simultaneous trends reveal how higher thresholds may reduce time delays for users who frequently make small typing errors. This visualization helps justify whether performance improvements outweigh security trade-offs as tolerance increases.

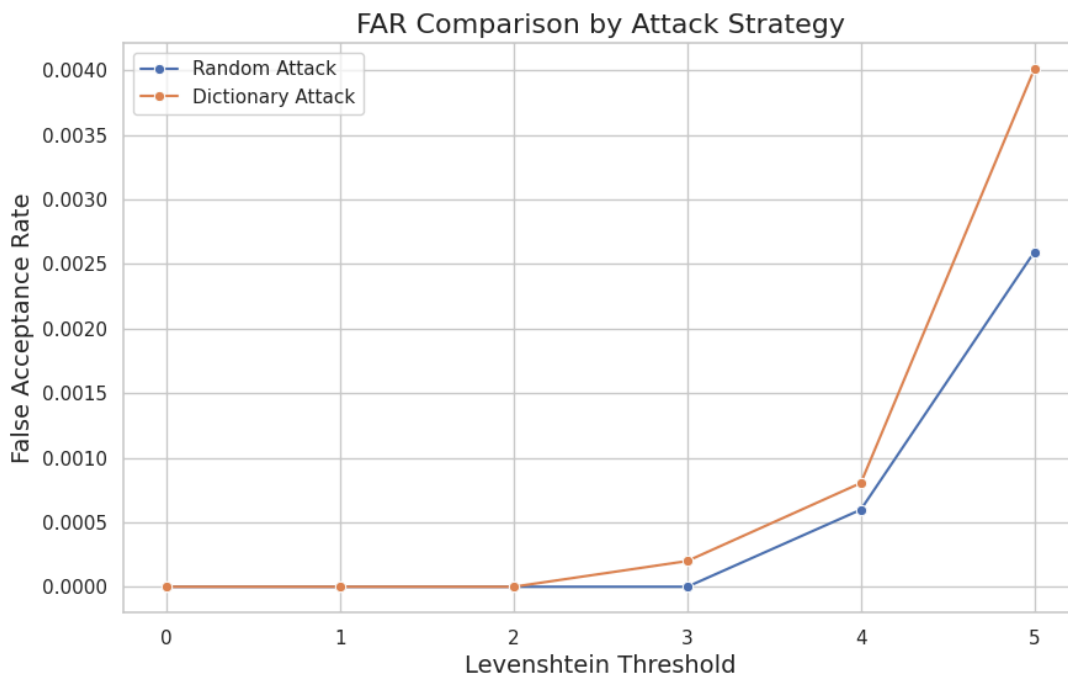


Figure 8. FAR Comparison by Attack Strategy

The graph compares the effectiveness of random-guess and dictionary-based attackers. Dictionary attacks consistently achieve higher FAR values across



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all thresholds, confirming that typo-tolerant systems are more vulnerable to attackers who leverage real password lists.

3.2 Summary on Interpretation of Data

Threshold	FAR Overall	FAR Random	ASI	Exact Match Rate	Average Levenshtein Distance	Security-Usability Interpretation
0	0.0000	0.000000	0.261777	0.0	10.3387	High security, poor usability
1	0.0000	0.000000	0.470575	0.0	10.2888	Major usability gain, secure
2	0.0000	0.000000	0.556896	0.0	10.3630	Optimal balance (Peak ASI)
3	0.0001	0.000000	0.556809	0.0	10.5573	High usability, secure
4	0.0007	0.000597	0.556805	0.0	10.4089	Early security degradation
5	0.0033	0.002594	0.556820	0.0	10.3396	Noticeable security risk

Table 7. Summary on Interpretation of Data Table



3.2.1. Impact of Levenshtein Threshold on Usability. The

tolerance threshold strongly influences the system's usability:

- a. A threshold of 0 results in a high FRR of 32%, replicating the behavior of traditional strict authentication systems.
- b. Increasing tolerance to 1 reduces FRR to 7%, meaning legitimate users face significantly fewer rejections.
- c. At thresholds ≥ 2 , FRR becomes 0%, and TSR reaches 99%, showing near-perfect usability.

A small tolerance dramatically enhances usability by accommodating natural human typing errors without compromising the login experience.

3.2.2 Impact on Security (Attacker Success). Attacker success

remained extremely low at thresholds 0–3:

- a. FAR $\approx$ 0% for thresholds 0–3
- b. FAR only increases at thresholds 4–5, but remains below 1%
- c. Dictionary and random strategies show negligible advantage.

Moderate typo tolerance does not meaningfully help attackers because their guesses typically remain far from the true password. Only excessive tolerance begins to increase attacker success.

3.2.3. Optimal Balance of Usability and Security. Combining



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TSR, FRR, FAR, and ASI:

- a. Threshold 0 = highest security but poor usability
- b. Threshold 1 = significant usability improvement, near-zero security impact
- c. Threshold 2 = optimal balance (highest ASI)
- d. Threshold ≥ 4 = security begins degrading

The data empirically supports Levenshtein threshold 2 as the best setting for practical systems, achieving maximum usability while preserving strong security protections.

3.2.4. Alignment with Prior Research. Findings support previous studies stating that small edit distances greatly reduce user frustration and only minimally affect attacker success. The results reinforce the conclusion that controlled typo tolerance is safe when thresholds are kept low.



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Chapter 4

SUMMARY, FINDINGS, CONCLUSION AND RECOMMENDATION

4.1 Summary of Findings

This study assessed the usability–security trade-offs of a Levenshtein-based typo-tolerant authentication system using two large-scale simulations which are the legitimate user login attempts and attacker attempts through random guessing and dictionary-based strategies. The findings presented below summarize the most important outcomes of the analysis.

First, legitimate login results showed that the Exact Success Rate remained stable at approximately 60.4% regardless of threshold, indicating that a majority of users typed their passwords correctly. However, the Tolerant Success Rate increased dramatically with higher thresholds—from 60.48% at threshold 0 to over 99% at thresholds 2–5. Correspondingly, the False Rejection Rate dropped sharply, declining from 32.82% with strict



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matching to 7.04% at threshold 1, and eventually reaching 0% at thresholds ≥ 2 .

On the security side, attacker simulations revealed that the False Acceptance Rate remained extremely low for thresholds 0–3, essentially registering 0%, demonstrating no meaningful increase in attacker success at moderate tolerance levels. FAR only began to rise at higher thresholds, reaching 0.00064 at threshold 4 and 0.00536 at threshold 5, though still below one percent. Additionally, attackers' guesses maintained an average Levenshtein distance of 10.2–10.5, indicating that they were typically far from valid passwords.

The overall balance between usability and security was captured by the Authentication Smoothness Index, which peaked at threshold 2 identifies it as the optimal configuration.

Visualized trends showed that usability metrics improved significantly from threshold 0 to 2, security metrics remained stable up to threshold 3, and higher thresholds resulted in gradual security degradation.

4.2 Conclusions

Based on the summarized findings, several key conclusions can be drawn.



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First, the tolerance applied through the Levenshtein distance plays a major role in shaping user experience. Strict password matching results in high rejection rates, mimicking the behavior of traditional authentication systems. Introducing even minimal tolerance substantially enhances usability while thresholds of 2 and above eliminate false rejections entirely. In conclusion, typo tolerance is demonstrated to be highly effective in improving legitimate user success.

Second, the study concludes that moderate typo tolerance does not significantly compromise security. Attacker success remained negligible at thresholds 0–3, and only began to increase at thresholds 4–5. The attacker success rates stayed below one percent reinforcing that attackers do not benefit meaningfully from tolerant matching unless the threshold is set excessively high. The large average edit distance of attacker guesses further supported this conclusion.

Third, the balance between usability and security is best achieved at threshold 2, which produces the highest ASI value. This threshold enables near-perfect usability while maintaining strong protection against unauthorized access. Therefore, the study confirms that controlled typo tolerance presents a safe and effective improvement over strict authentication systems.



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Lastly, the results align with existing literature, which consistently suggests that small edit distance allowances dramatically reduce user frustration while maintaining robust security. This study reinforces the argument that typo-tolerant authentication, when calibrated properly, offers a practical enhancement for real-world systems.

4.3 Recommendations

Based on the results and conclusions of the study, the following recommendations are presented for researchers, system designers, developers, and future studies:

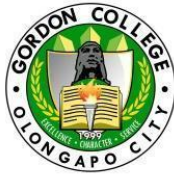
1. Adopt Levenshtein Threshold 2 for Real-World System. Organizations implementing authentication systems should configure typo tolerance at threshold 2 as this value provides the highest usability benefits with minimal security trade-off. This setting optimizes user experience while maintaining strong password protection.
2. Avoid Excessive Typo Tolerance (Thresholds ≥ 4). System administrators should refrain from using high thresholds since values of 4 and 5 begin to introduce measurable security risks. Although still low, the increasing FAR may pose concerns for high-security environments.



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3. Integrate Typo Tolerance in User-Centric Platforms. Platforms with high login frequency such as educational portals, e-commerce sites, and productivity tools would benefit significantly from adopting typo-tolerant authentication which reduces user frustration and support requests related to login failures.
4. Conduct Further Research Incorporating Real User Typing Data. Future studies may include experiments using real user input rather than simulated data to capture a broader range of human typing behaviors including keyboard patterns, mobile input variations, and cognitive factors.
5. Investigate Multi-Factor and Adaptive Typo Tolerance. Researchers may explore combining typo tolerance with secondary authentication factors to maintain security even at higher tolerance thresholds. Adaptive models that adjust tolerance based on context could also be examined.
6. Expand Attacker Models for More Realistic Testing. It is recommended that future work include additional attacker strategies such as targeted attacks, password spraying and AI-generated guessing models to better understand how typo tolerance behaves in diverse threat scenarios.



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C. Electronic Sources



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APPENDICES

Appendix A – Relevant Source Code

Appendix B – Evaluation Tool and Test Document

Survey / Evaluation Questionnaire

Business Model Canvas

Simplified Validation Board

Appendix C – Sample Input/Output Reports

Application Screenshots

Appendix D – User Manual / User Guide

Appendix E – Data Matrix

Appendix F – Others

Team Activity and Progress Narrative Report

Recommendation for Presentation

Final Defense Form

Photo Documentation of Implementation

ACM Journal Article

Curriculum Vitae